Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

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Instructions

1. Which procedural steps are integral to implementing the Naive Bayes algorithm for the task of spam filtering? Select all that apply. 1 / 1 point

- [ ] Employing gradient descent to minimize the associated loss function
- [x] Computing the posterior probabilities for each class

  Nice work

  Correct! The Naive Bayes algorithm relies on the calculation of posterior probabilities to classify the data into various categories.

- [x] Tokenizing the textual data

  Nice work

  Correct! Tokenization is a fundamental preprocessing step that transforms text into a format suitable for the Naive Bayes algorithm.

- [ ] Pruning the decision tree to mitigate overfitting
- [x] Transforming the text into a numerical representation

  Nice work

  Correct! Converting textual data into numerical vectors is essential for applying the Naive Bayes method effectively.

2. What represents a notable limitation when employing linear regression for the purpose of spam detection? 1 / 1 point

- ( ) Inapplicability to binary classification scenarios
- ( ) Exclusively requires categorical data
- ( ) Inability to process extensive datasets
- (•) Presumes a linear relationship between input features and the outcome variable

  Nice work

  Correct! Linear regression assumes a linear relationship between inputs and the target, which may not effectively capture the complex patterns necessary for accurate spam detection.

3. What technique is predominantly utilized to represent text data as numerical vectors for machine learning algorithms? 1 / 1 point

- (•) TF-IDF (Term Frequency-Inverse Document Frequency)
- ( ) Linear Discriminant Analysis
- ( ) K-means Clustering
- ( ) Principal Component Analysis

  Nice work

  Correct! TF-IDF is widely employed for converting textual information into numerical vectors, enhancing its usability in machine learning models.

4. Which attributes are advantageous when deploying decision tree algorithms for the purpose of spam filtering? Select all that apply. 1 / 1 point

- [x] Capable of handling both numerical and categorical data seamlessly

  Nice work

  Correct! Decision trees can effectively manage a combination of numerical and categorical features within the dataset.

- [ ] Vulnerable to overfitting
- [x] Adaptable to large-scale datasets

  Nice work

  Correct! Decision trees are scalable and can handle large datasets efficiently, enhancing their applicability in diverse scenarios.

- [x] Intuitive and straightforward to interpret

  Nice work

  Correct! Decision trees are prized for their clarity and ease of interpretation, making them user-friendly for analysis.

- [ ] Necessitates a substantial volume of data to achieve optimal performance

5. You are tasked with improving a spam detection system that currently relies on basic keyword matching. Which machine learning algorithm would provide the most effective balance of simplicity, performance, and ease of implementation for text classification tasks like spam filtering? 1 / 1 point

- ( ) Support Vector Machine
- ( ) Random Forest
- (•) Naive Bayes
- ( ) K-Nearest Neighbors

  Nice work

  Correct! Naive Bayes is highly efficient and effective for text classification, including spam detection, due to its simplicity and ease of implementation.